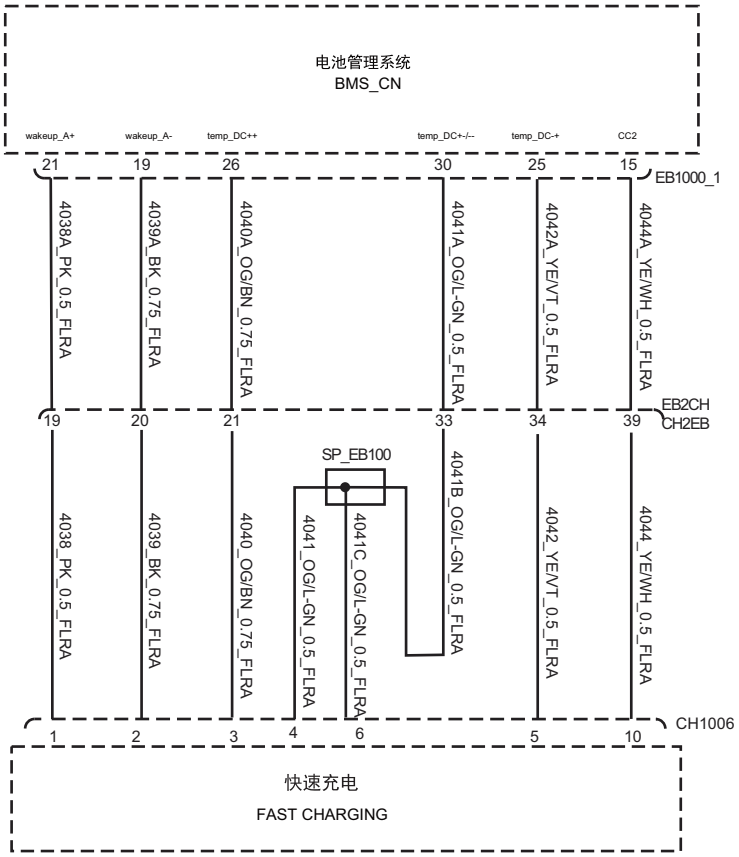
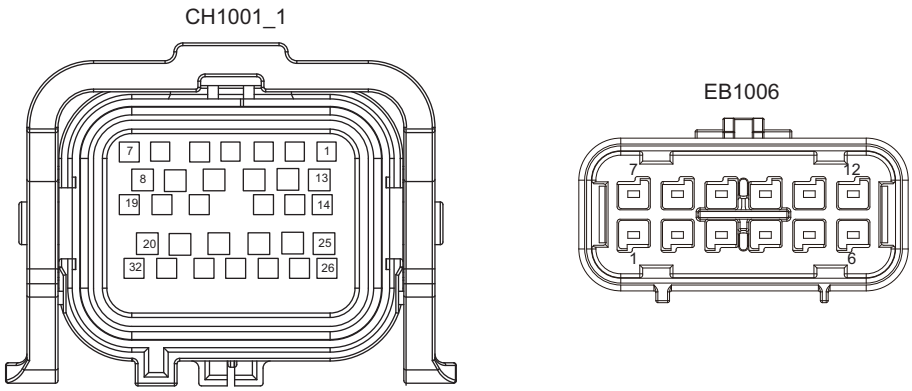
P1DF3-F1、 P1DF3-F0

Fault diagnosis code
P1DF3-F1：DC charging socket over temperature level 2
P1DF3-F2：DC charging socket over temperature level 1
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Temperature sensor fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1DF3-F1
DC charging socket temperature sampling value and valid and temperature ⩾ 120 ℃ &&DC charging gun insertion is effective
P1DF3-F0
DC charging socket temperature sampling value and valid and temperature ⩾ 100 ℃ &&DC charging gun insertion is effective
To set the effect of a fault code condition
Description of circuit diagram

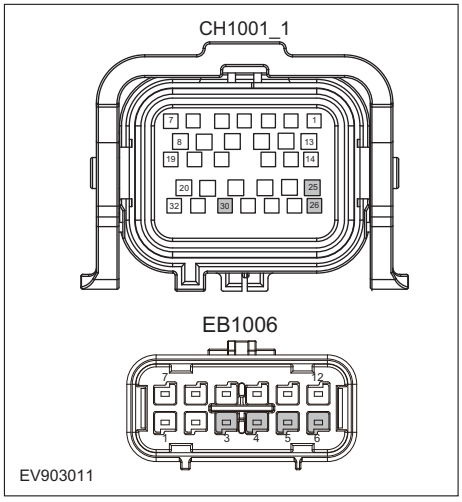
Circuit diagram

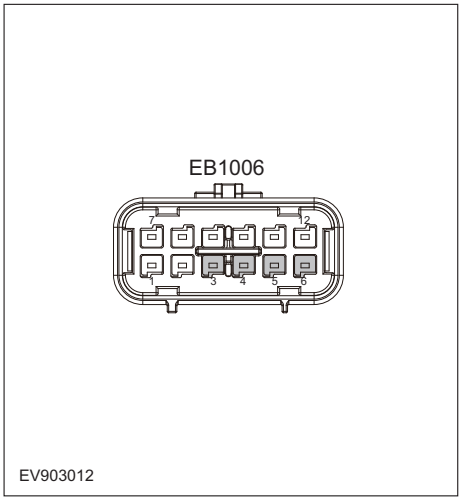




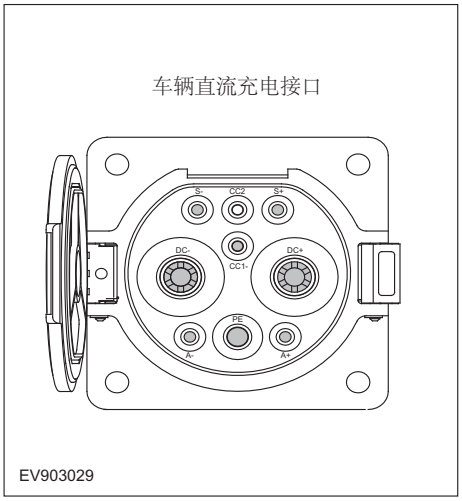
EV903010

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of BMS for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 If there is any fault code except for P1DF3-F1,P1DF3-F0?
		YES → Refer to:DTC Summary list(BMS).
NO ↓		
Step 3	Check CDU.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the CDU with diagnostic apparatus. 4 Test whether the faulted code is existed.
		YES → DTC diagnosis is performed based on specific fault codes.
NO ↓		
Step 4	Check the DC power supply.	
		1 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 2 Check the DC power supply. 3 Check whether the result is normal or not?
		NO → Replace the AC power in good condition.
YES ↓		

Step 5	Check the circuit between fast charging and BMS. (Open circuit)
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1006 of fast charging. 3 Disconnect the harness plug EB1001_1 of BMS. 4 Measure the resistance between the harness terminals of fast charging and BMS. Measuring circuit: resistance between terminal 3 on plug EB1006 and terminal 26 on plug EB1001_1. Measuring circuit: resistance between terminal 4 on plug EB1006 and terminal 30 on plug EB1001_1. Measuring circuit: resistance between terminal 5 on plug EB1006 and terminal 25 on plug EB1001_1. Measuring circuit: resistance between terminal 6 on plug EB1006 and terminal 30 on plug EB1001_1. **Standard value: < 1Ω** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the circuit of fast charging. (Short circuit)
	1 Turn the ignition switch to OFF. 2 Connect the harness plug EB1001_1 of BMS. 3 Measure the resistance between the harness terminal of fast charging and ground. Measuring circuit: resistance between terminal 3 on plug EB1006 and ground. Measuring circuit: resistance between terminal 4 on plug EB1006 and ground. Measuring circuit: resistance between terminal 5 on plug EB1006 and ground. Measuring circuit: resistance between terminal 6 on plug EB1006 and ground. **Standard value: ∞** 4 Measure the voltage between the harness terminal of BMS and ground. Measuring circuit: voltage between terminal 3 on plug EB1006 and ground. Measuring circuit: voltage between terminal 4 on plug EB1006 and ground. Measuring circuit: voltage between terminal 5 on plug EB1006 and ground. Measuring circuit: voltage between terminal 6 on plug EB1006 and ground. **Standard value: ≈ 0V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the DC charging gun （functional check).
 EV903028	1 Measure the resistance between the harness terminals of DC charging gun. Measure circuit: resistance between the DC+ and DC- terminals of the DC charging gun. **Standard value: ∞** Measure circuit: resistance between A+ and A- terminals of the DC charging gun. **Standard value: ∞** Measure circuit: resistance between the S+ and S- terminals of the DC charging gun. **Standard value: 120 Ω** Measure circuit: resistance between CC1 and CC2 terminals of the DC charging gun. **Standard value: ∞** Measure circuit: resistance between the CC1 and PE terminals of the DC charging gun. **Standard value(connect the confirmation switch): ∞** **Standard value(disconnect the confirmation switch): 1000 Ω** Measure circuit: resistance between the CC2 and PE terminals of the DC charging gun. **Standard value: 1000 Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Check the DC charging port of the vehicle （functional check).
	1 Turn the ignition switch to OFF. 2 Open the vehicle DC charging port cover. 3 Measure the resistance between the harness terminals of DC charging port. Measure circuit: resistance between the DC+ and DC- terminals of the DC charging gun. **Standard value: ∞** Measure circuit: resistance between A+ and A- terminals of the DC charging gun. **Standard value: ∞** Measure circuit: resistance between the S+ and S- terminals of the DC charging gun. **Standard value: 120 Ω** Measure circuit: resistance between the CC1 and PE terminals of the DC charging gun. **Standard value: 1000 Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 9	Check the high-voltage cable connecting the DC charging port of the vehicle to the CDU (wiring harness conduction check).
	1 Turn the ignition switch to OFF. 2 Disconnect AC charging high-voltage cable of CDU. 3 Measure the resistance between the harness terminals of DC charging port interface and plug. Measure circuit: the resistance between the DC+,, DC- of AC charging port and CDU. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 10	Test whether BMS is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1000_1 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BMS.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	